Architecture Constraint Document (ACD) — Version 1

System: Quantitative Algorithmic Trading System (India / SEBI-regulated) **Phase:** Pre-Architecture — Constraint Discovery **Status:** FINAL — Constraint Extraction Complete **Source Corpus:** 6 files — 01_architecture_driver_registry, 05_root_constraint_registry, 05_adr_candidate_corpus, 06_adr_quality_audit, 06_architecture_core, 09_architecture_readiness_reassessment **Total ADRs Surveyed:** 120 **Architecture Readiness Score:** 85/100 — Architecture planning BLOCKED (17 unresolved blockers) **Date:** 2026-06-04

> **Scope Declaration** This document contains only what must be true regardless of architecture. No technologies are recommended. No databases are specified. No frameworks are proposed. No implementation plans are contained herein. Every constraint below remains true even if every technology, database, model, and deployment method changes.

Constraint Classification Legend

Code	Category
REG	Regulatory
REL	Reliability
GOV	Governance
VAL	Validation
DAT	Data
HUM	Human Oversight
SEC	Security
AIB	AI Boundary
OPS	Operational

Constraint Confidence Legend

Level	Meaning
HIGH	STRONG evidence, direct regulatory mandate, or cascading collapse on violation
MEDIUM	MODERATE evidence, confirmed failure mode, no opposing evidence
LOW	Inferred constraint, documented assumption, awaiting empirical validation

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SECTION 1 — REGULATORY CONSTRAINTS

ACD-REG-001

Constraint Statement: All order routing must comply with SEBI-mandated rate limits. The system must not exceed the maximum allowable order submission frequency imposed by SEBI regulation regardless of strategy throughput demands or infrastructure scaling decisions.

Source Files: 01_architecture_driver_registry (DRV-007), 05_root_constraint_registry (ADR-008), 06_architecture_core (ADR-008), 05_adr_candidate_corpus (ADR-008)

Supporting ADRs: ADR-008, ADR-086

Problem Prevented: Regulatory breach of SEBI API rate limits leading to static IP blacklisting, account suspension, or broker API bans.

Failure If Violated: Account suspension under SEBI enforcement. Permanent static IP block. Loss of market access. Trade execution daemon shutdown.

Criticality: CRITICAL

Confidence: HIGH

Category: REG

ACD-REG-002

Constraint Statement: All broker API connections must authenticate using the authentication mechanism mandated by SEBI. Authentication must be cryptographically verified on each session. Manual credential bypass is not permissible in any production pathway.

Source Files: 05_root_constraint_registry (ADR-008), 06_architecture_core (ADR-008), 05_adr_candidate_corpus (ADR-008, ADR-087)

Supporting ADRs: ADR-008, ADR-087, ADR-117

Problem Prevented: Unauthenticated or session-spoofed order routing in violation of SEBI OAuth mandate.

Failure If Violated: Regulatory non-compliance. Broker API connection rejection. Full trade execution halt.

Criticality: CRITICAL

Confidence: HIGH

Category: REG, SEC

ACD-REG-003

Constraint Statement: All production API connections to regulated broker endpoints must originate from a fixed, registered network address. Dynamic or rotating network addressing is prohibited on any pathway that transmits orders to an exchange.

Source Files: 05_root_constraint_registry (ADR-008), 06_architecture_core (ADR-008)

Supporting ADRs: ADR-008, ADR-086

Problem Prevented: SEBI static IP mandate violation. Broker session rejection on dynamic IP detection.

Failure If Violated: Connection rejection by broker. Account flagged for regulatory review. Trade execution halt.

Criticality: CRITICAL

Confidence: HIGH

Category: REG, SEC

ACD-REG-004

Constraint Statement: All historical price data used in backtesting must be adjusted for corporate actions including stock splits, dividends, and mergers. Unadjusted data must not be used as a sole or primary backtesting input under any configuration.

Source Files: 05_adr_candidate_corpus (ADR-005, ADR-010), 01_architecture_driver_registry (DRV-004), 06_adr_quality_audit

Supporting ADRs: ADR-005, ADR-010

Problem Prevented: Backtests producing false performance metrics due to price discontinuities at split and dividend events. Strategy deployment decisions made on corrupted historical signals.

Failure If Violated: Strategies validated on incorrect historical prices deployed to live capital. Capital destruction through corrupted signal logic.

Criticality: HIGH

Confidence: HIGH

Category: REG, DAT, VAL

SECTION 2 — DATA CONSTRAINTS

ACD-DAT-001

Constraint Statement: Market data used for signal generation must be validated against a minimum of two independent data providers before any signal logic executes. Single-source data pipelines are prohibited from feeding execution-bound signal logic.

Source Files: 01_architecture_driver_registry (DRV-004), 06_architecture_core (ADR-011), 05_adr_candidate_corpus (ADR-011), 05_root_constraint_registry (ADR-011)

Supporting ADRs: ADR-004, ADR-011

Problem Prevented: Silent data corruption, dropped candles, and provider-specific pricing anomalies propagating into live signal logic unchallenged.

Failure If Violated: Corrupted signals execute in live market. Capital destruction via stale or erroneous pricing. Incorrect position sizing from bad OHLCV data.

Criticality: HIGH

Confidence: HIGH

Category: DAT, VAL

ACD-DAT-002

Constraint Statement: Intraday time-series feeds must be verified for completeness at ingestion. Any feed that contains gaps, dropped intervals, or missing candles must be flagged and quarantined before reaching signal computation. Incomplete feeds must not be silently forward-filled or interpolated without explicit detection logging.

Source Files: 01_architecture_driver_registry (DRV-004), 06_architecture_core (ADR-012), 05_adr_candidate_corpus (ADR-012, ADR-040)

Supporting ADRs: ADR-012, ADR-040

Problem Prevented: Undetected feed gaps producing false indicators, phantom signal triggers, and backtesting results that do not reflect live data reality.

Failure If Violated: Silent feed corruption propagates through signal stack. Indicators computed on incomplete data produce erroneous trade signals in live execution.

Criticality: HIGH

Confidence: HIGH

Category: DAT, VAL

ACD-DAT-003

Constraint Statement: Backtesting datasets must include delisted instruments for the full backtesting period. Survivorship-biased datasets that exclude delisted stocks are prohibited as a sole backtesting source.

Source Files: 05_adr_candidate_corpus (ADR-013), 05_root_constraint_registry (ADR-013), 06_adr_quality_audit (ADR-013)

Supporting ADRs: ADR-013

Problem Prevented: Survivorship bias inflating historical strategy performance metrics, producing deployment decisions based on non-representative historical universes.

Failure If Violated: Strategy overfitting to stocks that survived. Live performance materially worse than backtest. Capital allocation based on fraudulent historical performance.

Criticality: HIGH

Confidence: MEDIUM

Category: DAT, VAL

ACD-DAT-004

Constraint Statement: Incorrect or anomalous pricing data must trigger an automatic halt to all downstream processing that depends on it. The system must not permit a bad pricing input to propagate unchecked through signal computation, order sizing, or order routing pipelines.

Source Files: 05_adr_candidate_corpus (ADR-017, ADR-118), 06_architecture_core (ADR-016), 06_adr_quality_audit (ADR-017)

Supporting ADRs: ADR-016, ADR-017, ADR-118

Problem Prevented: Infinite processing loops triggered by malformed pricing data. Runaway order routing from misinterpreted price inputs.

Failure If Violated: Unbounded compute loops consume system resources. Runaway orders submitted to exchange at incorrect prices. System crash under self-reinforcing bad data propagation.

Criticality: CRITICAL

Confidence: HIGH

Category: DAT, REL, OPS

ACD-DAT-005

Constraint Statement: Market data storage must be partitioned by time dimension at ingestion. Flat, unpartitioned, or document-oriented storage formats are prohibited for market data that will be queried analytically.

Source Files: 06_architecture_core (ADR-003, ADR-078), 05_adr_candidate_corpus (ADR-003), 05_root_constraint_registry (ADR-003)

Supporting ADRs: ADR-003, ADR-078

Problem Prevented: Query performance degradation on multi-year data scans. Memory exhaustion from full-scan reads on unpartitioned stores. Analytical pipeline timeouts during signal computation.

Failure If Violated: Analytical queries time out during live signal computation. Memory overflow halts process during high-frequency data reads.

Criticality: HIGH

Confidence: HIGH

Category: DAT, REL

ACD-DAT-006

Constraint Statement: Transaction and trade state records must preserve write-ahead integrity such that a hard system failure (power loss, crash) does not corrupt committed trade state. Any transaction storage mechanism must be recoverable to a consistent pre-crash state without manual intervention.

Source Files: 05_adr_candidate_corpus (ADR-015, ADR-020, ADR-043, ADR-079), 06_architecture_core (ADR-015)

Supporting ADRs: ADR-015, ADR-020, ADR-043, ADR-079

Problem Prevented: Transaction state corruption on unclean shutdown. Recovery to inconsistent position or P&L state after hardware failure.

Failure If Violated: System recovers to incorrect position state. Trades re-submitted or missed on restart. Capital loss from phantom or duplicate positions.

Criticality: CRITICAL

Confidence: MEDIUM

Category: DAT, REL

ACD-DAT-007

Constraint Statement: The data validation pipeline must be proven against synthetic anomaly injection before it is trusted to gate live signal computation. The pipeline must demonstrably detect, classify, and quarantine known anomaly patterns including gap fills, pricing spikes, and malformed records.

Source Files: 06_architecture_core (ADR-016), 05_adr_candidate_corpus (ADR-016), 05_root_constraint_registry (ADR-016)

Supporting ADRs: ADR-016

Problem Prevented: Silent validation failures where the pipeline appears functional but passes corrupted data. Undiscovered detection gaps in the anomaly classifier.

Failure If Violated: Data validation pipeline passes corrupted data silently. Live signals generated from anomaly-contaminated inputs. Backtest validation results are unreliable.

Criticality: HIGH

Confidence: HIGH

Category: DAT, VAL

ACD-DAT-008

Constraint Statement: No single external data provider may be designated as the sole authoritative source for any data type that feeds execution-bound logic. All execution-bound data pipelines must have a documented fallback or cross-verification path.

Source Files: 01_architecture_driver_registry (DRV-004), 05_adr_candidate_corpus (ADR-004, ADR-006), 06_adr_quality_audit (ADR-004)

Supporting ADRs: ADR-004, ADR-006, ADR-011

Problem Prevented: Single-provider outages, silent API changes, and structural data incompleteness propagating into live execution without detection.

Failure If Violated: Provider outage halts the entire data pipeline with no fallback. Silent API changes corrupt data without cross-validation detection.

Criticality: HIGH

Confidence: HIGH

Category: DAT, REL

ACD-DAT-009

Constraint Statement: Exchange-originated trade cancellation and clearinghouse void events must be explicitly handled in the transaction record. Retroactive trade erasures must not be silently absorbed as committed state. Capital buffer must exist to absorb retroactive cancellation exposure.

Source Files: 05_adr_candidate_corpus (ADR-019, ADR-020, ADR-114), 06_architecture_core (ADR-019)

Supporting ADRs: ADR-019, ADR-020, ADR-114

Problem Prevented: Committed position records diverging from actual cleared position due to unhandled exchange cancellation events. Incorrect P&L and margin calculations from phantom trades.

Failure If Violated: Phantom positions recorded as committed. Incorrect margin utilization. Capital loss from positions believed closed that were voided.

Criticality: CRITICAL

Confidence: MEDIUM

Category: DAT, REL, GOV

SECTION 3 — VALIDATION CONSTRAINTS

ACD-VAL-001

Constraint Statement: No strategy may be deployed to live capital unless it has passed a walk-forward validation sequence. K-fold or randomized cross-validation methodologies that violate temporal ordering of market data are prohibited as the primary deployment gate.

Source Files: 05_adr_candidate_corpus (ADR-007, ADR-027), 05_root_constraint_registry (ADR-007, ADR-027)

Supporting ADRs: ADR-007, ADR-027

Problem Prevented: Data leakage from future data into training windows. In-sample overfitting disguised as out-of-sample performance. Deployment of strategies whose live performance cannot be predicted from backtest results.

Failure If Violated: Strategies appear profitable in validation but fail immediately in live markets. Capital deployed to curve-fitted strategies. Catastrophic drawdown on live deployment.

Criticality: CRITICAL

Confidence: HIGH

Category: VAL

ACD-VAL-002

Constraint Statement: Strategy deployment must clear a minimum statistical significance threshold that accounts for multiple testing and data mining bias. The deployment threshold must be more conservative than standard 95% confidence intervals. Strategies that pass only a t-statistic of 2.0 must not be deployed to live capital.

Source Files: 05_adr_candidate_corpus (ADR-028, ADR-029), 05_root_constraint_registry (ADR-028, ADR-029)

Supporting ADRs: ADR-028, ADR-029

Problem Prevented: Deployment of spurious statistical artifacts as valid strategies. False discovery from data mining over large strategy search spaces.

Failure If Violated: Strategies with no genuine alpha deployed to live capital. Capital loss from noise-fitted strategies.

Criticality: HIGH

Confidence: HIGH

Category: VAL

ACD-VAL-003

Constraint Statement: Every production binary must be verified for hash integrity against a known-good reference before it is permitted to route live orders. Partial binary rollouts where nodes run different versions simultaneously are prohibited.

Source Files: 05_adr_candidate_corpus (ADR-030, ADR-054, ADR-081, ADR-083), 06_architecture_core (ADR-054), 01_architecture_driver_registry (DRV-005)

Supporting ADRs: ADR-030, ADR-054, ADR-081, ADR-083, ADR-105

Problem Prevented: Knight Capital-class failure from mixed-version binary deployment. Deprecated or untested code routing live orders on partial rollout nodes.

Failure If Violated: Runaway order routing from stale binary on unverified node. Catastrophic capital loss within minutes of incorrect binary activation.

Criticality: CRITICAL

Confidence: HIGH

Category: VAL, OPS, GOV

ACD-VAL-004

Constraint Statement: All configuration flags must be audited before any deployment to ensure no deprecated or legacy flag namespace can be reactivated. Configuration flag namespaces from prior versions must be permanently purged, not commented out or conditionally disabled.

Source Files: 06_architecture_core (ADR-031), 05_adr_candidate_corpus (ADR-031, ADR-082, ADR-108), 01_architecture_driver_registry (DRV-005), 05_root_constraint_registry (ADR-031)

Supporting ADRs: ADR-031, ADR-082, ADR-108

Problem Prevented: Knight Capital failure mode: legacy configuration flag reactivated by a deployment step, enabling a deprecated execution pathway on a production node.

Failure If Violated: Deprecated execution mode reactivated silently. Uncontrolled order flow from legacy pathway. Catastrophic capital loss.

Criticality: CRITICAL

Confidence: HIGH

Category: VAL, OPS

ACD-VAL-005

Constraint Statement: All production nodes must pass automated chaos and sanity testing before being permitted to route live orders. Node promotion to live routing without passing a defined automated test suite is prohibited.

Source Files: 05_adr_candidate_corpus (ADR-033, ADR-111), 05_root_constraint_registry (ADR-033)

Supporting ADRs: ADR-033, ADR-111

Problem Prevented: Silent node misconfiguration surviving to live routing. Undiscovered failure modes in execution paths not exercised during normal operation.

Failure If Violated: Misconfigured node routes orders with untested execution logic. Failure modes discovered in live capital exposure rather than pre-production testing.

Criticality: HIGH

Confidence: HIGH

Category: VAL, OPS

ACD-VAL-006

Constraint Statement: The system must continuously monitor for distributional shift between live market data and training data. When shift exceeds a defined threshold, execution must be halted or position sizing must be reduced automatically. Continued trading through undetected regime change is prohibited.

Source Files: 06_architecture_core (ADR-034), 05_adr_candidate_corpus (ADR-034, ADR-048, ADR-063), 01_architecture_driver_registry (DRV-006)

Supporting ADRs: ADR-034, ADR-048, ADR-063

Problem Prevented: Strategy operating in a market regime it was never validated for. Model assumptions violated silently during regime shift. Undetected concept drift producing increasing losses with no automatic response.

Failure If Violated: Strategy continues executing through fundamental regime change. Model performance collapses. Losses accumulate without triggering any protective response.

Criticality: HIGH

Confidence: HIGH

Category: VAL, REL

ACD-VAL-007

Constraint Statement: Execution loops must contain hard-coded, synchronous checks that verify parent order balance state before any child order is submitted. Balance verification must be inseparable from the execution loop — it cannot be implemented as an asynchronous, optional, or out-of-band check.

Source Files: 06_architecture_core (ADR-032), 05_adr_candidate_corpus (ADR-032, ADR-107), 01_architecture_driver_registry (DRV-006)

Supporting ADRs: ADR-032, ADR-107

Problem Prevented: Runaway child orders submitted beyond parent order capacity. Execution loops continuing after parent balance exhaustion.

Failure If Violated: Unbounded child order submission. Position sizes exceed intended parent allocation. Capital overexposure from runaway loop execution.

Criticality: CRITICAL

Confidence: HIGH

Category: VAL, REL

SECTION 4 — RELIABILITY CONSTRAINTS

ACD-REL-001

Constraint Statement: The system must implement circuit breakers that automatically disconnect execution from market data feeds when consolidated tape latency exceeds a defined threshold. Continued order routing on stale market data is prohibited.

Source Files: 05_adr_candidate_corpus (ADR-037, ADR-089, ADR-112), 01_architecture_driver_registry (DRV-006)

Supporting ADRs: ADR-034, ADR-037, ADR-089, ADR-112

Problem Prevented: Stale data driving execution decisions. Orders routed on prices that no longer reflect current market state. Adverse fills from latency-induced execution on outdated quotes.

Failure If Violated: Orders submitted at stale prices. Execution at significantly off-market quotes. Capital loss from latency-exploited fills.

Criticality: CRITICAL

Confidence: HIGH

Category: REL, OPS

ACD-REL-002

Constraint Statement: The system must monitor order book depth in real time and must halt market order submission when order book depth falls below a defined minimum. Market orders must not be routed into an evaporated order book.

Source Files: 05_adr_candidate_corpus (ADR-038, ADR-113), 01_architecture_driver_registry (DRV-006)

Supporting ADRs: ADR-038, ADR-113

Problem Prevented: Market impact self-amplification during thin book conditions. Catastrophic slippage from routing market orders into illiquid conditions.

Failure If Violated: Market orders consume entire visible book depth. Extreme slippage. Price dislocation accelerated by system's own order flow. Capital destruction from self-inflicted market impact.

Criticality: CRITICAL

Confidence: HIGH

Category: REL, OPS

ACD-REL-003

Constraint Statement: Position sizing must automatically scale down when model uncertainty exceeds a defined threshold. The system must treat high model uncertainty as a direct input to order sizing — not merely a logging event.

Source Files: 05_adr_candidate_corpus (ADR-035, ADR-097), 01_architecture_driver_registry (DRV-006)

Supporting ADRs: ADR-035, ADR-097

Problem Prevented: Full-size orders submitted under high uncertainty conditions. Maximum capital exposure deployed precisely when model confidence is lowest.

Failure If Violated: Maximum position sizes deployed into highest-uncertainty market conditions. Expected value of trade becomes negative under model uncertainty premium.

Criticality: CRITICAL

Confidence: MEDIUM

Category: REL, VAL

ACD-REL-004

Constraint Statement: Order volume must be bounded relative to recent market volume. The system must not submit orders that represent more than a defined maximum percentage of trailing market volume in any given interval.

Source Files: 05_adr_candidate_corpus (ADR-036)

Supporting ADRs: ADR-036

Problem Prevented: System's own order flow creating self-reinforcing price impact. Market manipulation regulatory risk from volume-dominant order submission.

Failure If Violated: System orders move market price adversely. Execution at systematically worse-than-expected prices. Regulatory inquiry into volume concentration.

Criticality: HIGH

Confidence: HIGH

Category: REL, REG

ACD-REL-005

Constraint Statement: The system must implement deterministic, bounded retry logic for all rate-limited API responses. Unbounded retry loops and immediate retry-on-rejection are prohibited. The retry interval must increase in a predictable pattern and must have a defined maximum attempt ceiling.

Source Files: 05_adr_candidate_corpus (ADR-085, ADR-106), 05_root_constraint_registry (ADR-085)

Supporting ADRs: ADR-085, ADR-106

Problem Prevented: Retry storms on rate-limited endpoints. HTTP 429 responses triggering rapid-fire retries that exhaust rate limit budgets and risk shadow IP bans from brokers.

Failure If Violated: Retry storm consumes entire rate limit budget. Shadow IP ban from broker on excessive 429 response pattern. Full trading halt from IP-level block.

Criticality: HIGH

Confidence: HIGH

Category: REL, REG, OPS

ACD-REL-006

Constraint Statement: Bespoke risk scenario limits must be enforced by hard caps that cannot be overridden by individual operators or client-facing personnel. The limit override process must require multi-party authorization and must leave an immutable audit record.

Source Files: 06_architecture_core (ADR-054), 05_adr_candidate_corpus (ADR-054), 01_architecture_driver_registry (DRV-006)

Supporting ADRs: ADR-054, ADR-073

Problem Prevented: Risk manager inflation of scenario limits for client accommodation. Unauthorized expansion of risk parameters outside governance process.

Failure If Violated: Individual override of risk caps without institutional approval. Concentrated exposure exceeding governance limits. Systemic risk event from uncapped scenario escalation.

Criticality: HIGH

Confidence: HIGH

Category: REL, GOV, HUM

SECTION 5 — GOVERNANCE CONSTRAINTS

ACD-GOV-001

Constraint Statement: All deprecated production code must be permanently removed from the deployable codebase before a binary is promoted to production. Deprecated code that is conditionally disabled, commented out, or gated by a flag that could be reactivated is not considered removed.

Source Files: 05_adr_candidate_corpus (ADR-009, ADR-108), 05_root_constraint_registry (ADR-009), 01_architecture_driver_registry (DRV-007)

Supporting ADRs: ADR-009, ADR-031, ADR-108

Problem Prevented: Knight Capital failure class: conditional reactivation of deprecated execution pathways during deployment. Legacy code surviving as a latent risk vector.

Failure If Violated: Deprecated code reactivated on one node during partial deployment. Uncontrolled legacy execution pathway routes orders. Catastrophic unrecoverable capital loss.

Criticality: CRITICAL

Confidence: HIGH

Category: GOV, OPS, VAL

ACD-GOV-002

Constraint Statement: Risk committee oversight must be active and deliberate. Passive, asynchronous, or email-only approval processes do not satisfy the governance requirement. High-impact risk decisions require synchronous, in-session review by the risk committee.

Source Files: 06_architecture_core (ADR-073), 05_adr_candidate_corpus (ADR-073, ADR-064), 01_architecture_driver_registry (DRV-003), 09_architecture_readiness_reassessment

Supporting ADRs: ADR-073, ADR-064, ADR-077

Problem Prevented: Rubber-stamp governance where risk decisions are approved without genuine deliberation. Committee members approving changes without reviewing actual risk exposure.

Failure If Violated: High-impact risk changes approved without deliberate review. Governance process becomes a formality. Risk escalation paths not properly activated during crisis.

Criticality: HIGH

Confidence: MEDIUM

Category: GOV, HUM

ACD-GOV-003

Constraint Statement: All production deployments must follow a coordinated promotion process across all nodes simultaneously. No partial or staggered binary rollout is permitted where live routing nodes run different binary versions concurrently.

Source Files: 05_adr_candidate_corpus (ADR-083), 05_root_constraint_registry (ADR-083)

Supporting ADRs: ADR-030, ADR-083

Problem Prevented: Mixed-version execution across nodes producing inconsistent order routing behavior. Version mismatch creating unpredictable execution outcomes on multi-node systems.

Failure If Violated: Different nodes execute orders under different logic simultaneously. Inconsistent fills, duplicate orders, or missed hedges from version divergence.

Criticality: CRITICAL

Confidence: HIGH

Category: GOV, OPS, VAL

SECTION 6 — HUMAN OVERSIGHT CONSTRAINTS

ACD-HUM-001

Constraint Statement: Execution of financial orders must remain under human gate control. No execution pathway may route a live order to an exchange without a human approval event in the causal chain, regardless of the degree of automation in signal generation or risk computation.

Source Files: 06_architecture_core (ADR-071), 05_adr_candidate_corpus (ADR-071, ADR-096), 01_architecture_driver_registry (DRV-002), 05_root_constraint_registry, 09_architecture_readiness_reassessment

Supporting ADRs: ADR-071, ADR-096

Problem Prevented: Non-deterministic autonomous execution triggered by AI reasoning errors, hallucinations, or unexpected model output formatting. Full autonomy loop with no human abort path.

Failure If Violated: AI model directly routes orders without human confirmation. Non-deterministic or hallucinated order parameters submitted to exchange. Unrecoverable capital loss from autonomous AI execution error.

Criticality: CRITICAL

Confidence: HIGH

Category: HUM, AIB, REG

ACD-HUM-002

Constraint Statement: Critical alert systems must be governed by documented runbooks that are maintained, tested, and distinct from general operational runbooks. Alert fatigue mitigation is a design requirement — the alert system must be calibrated to sustain operator attention on genuinely critical events.

Source Files: 06_architecture_core (ADR-072), 05_adr_candidate_corpus (ADR-072, ADR-109), 01_architecture_driver_registry (DRV-003), 09_architecture_readiness_reassessment

Supporting ADRs: ADR-072, ADR-109

Problem Prevented: Operator desensitization from alert flood. Critical alerts missed because operators have habituated to undifferentiated alert noise.

Failure If Violated: Operators miss critical execution or risk alerts. Genuine emergency not escalated. Risk event escalates beyond recoverable threshold while humans are not responding.

Criticality: HIGH

Confidence: MEDIUM

Category: HUM, OPS

ACD-HUM-003

Constraint Statement: Multi-signature authorization is required before any position limit override or risk parameter expansion beyond baseline. No single operator may authorize a limit override unilaterally.

Source Files: 05_adr_candidate_corpus (ADR-073), 01_architecture_driver_registry (DRV-003), 06_architecture_core (ADR-073)

Supporting ADRs: ADR-073, ADR-054

Problem Prevented: Unilateral limit inflation by a single operator under client or time pressure. Social engineering attacks on operators to expand risk parameters.

Failure If Violated: Single operator inflates risk limits without institutional approval. Exposure exceeds authorized capital bounds. Governance failure enables concentrated loss event.

Criticality: HIGH

Confidence: MEDIUM

Category: HUM, GOV

ACD-HUM-004

Constraint Statement: Autonomous growth of trade volume beyond authorized parameters is prohibited under all circumstances. Any pathway that would allow the system to self-authorize volume expansion — including AI-inferred scaling — must be physically blocked, not merely policy-prohibited.

Source Files: 05_adr_candidate_corpus (ADR-076), 09_architecture_readiness_reassessment

Supporting ADRs: ADR-076, ADR-071

Problem Prevented: AI or algorithmic self-scaling of volume beyond human-authorized bounds. Gradual autonomous volume creep that circumvents position limits.

Failure If Violated: System executes orders at multiples of authorized volume. Undetected volume drift until drawdown event. Regulatory exposure from unauthorized volume concentration.

Criticality: HIGH

Confidence: MEDIUM

Category: HUM, AIB, GOV

SECTION 7 — AI BOUNDARY CONSTRAINTS

ACD-AIB-001

Constraint Statement: AI or language model components must be isolated exclusively to research and analytical functions. No AI component may have a direct or indirect pathway to execution APIs, order routing logic, or any system component that can submit orders to an exchange.

Source Files: 06_architecture_core (ADR-071, ADR-078), 05_adr_candidate_corpus (ADR-071, ADR-093, ADR-094), 01_architecture_driver_registry (DRV-002)

Supporting ADRs: ADR-071, ADR-078, ADR-093, ADR-094, ADR-095

Problem Prevented: Non-deterministic AI output directly routing orders. AI hallucinated parameters (price, quantity, direction) passed to order execution without human interception.

Failure If Violated: AI submits orders without human gate. Hallucinated order parameters executed at exchange. Immediate, potentially unrecoverable capital loss.

Criticality: CRITICAL

Confidence: HIGH

Category: AIB, HUM, REG

ACD-AIB-002

Constraint Statement: AI models must not be used for deterministic sorting, binary arithmetic, or any computation where exact numeric precision is required by execution logic. These computations must be performed by deterministic code that does not involve a language model inference step.

Source Files: 05_adr_candidate_corpus (ADR-099), 05_root_constraint_registry (ADR-099)

Supporting ADRs: ADR-099

Problem Prevented: Language model numeric errors producing catastrophically incorrect execution parameters. AI rounding, hallucination, or context-length degradation corrupting exact arithmetic required for trade sizing or ordering.

Failure If Violated: Incorrect binary formatting or sort ordering of execution parameters. Orders submitted with malformed quantities or wrong price precision. Exchange rejects or accepts malformed orders with unintended consequences.

Criticality: CRITICAL

Confidence: HIGH

Category: AIB, VAL

ACD-AIB-003

Constraint Statement: AI models must not be used to auto-generate, auto-validate, or auto-deploy backtesting code. Backtesting code must be written, reviewed, and validated by deterministic, human-auditable processes. AI-generated backtesting logic may introduce survivorship bias or curve-fitting that is not detectable without human review.

Source Files: 05_adr_candidate_corpus (ADR-100), 05_root_constraint_registry (ADR-100)

Supporting ADRs: ADR-100

Problem Prevented: AI-generated backtesting code producing results that are systematically biased toward positive outcomes. Curve-fitted strategies passed through AI-generated validation pipelines that are themselves optimized to produce favorable results.

Failure If Violated: Backtests report false alpha. Strategies with no live edge deployed to capital. Systematic overfitting masked by AI-generated validation.

Criticality: HIGH

Confidence: HIGH

Category: AIB, VAL

ACD-AIB-004

Constraint Statement: AI language model output must not be passed directly as query input to any analytical data store without a validation and sanitization layer. Dynamic query generation by a language model against live data stores is prohibited.

Source Files: 05_adr_candidate_corpus (ADR-021, ADR-101), 05_root_constraint_registry (ADR-021, ADR-101), 06_adr_quality_audit (ADR-021)

Supporting ADRs: ADR-021, ADR-101

Problem Prevented: Hallucinated queries causing out-of-memory events, table corruption, or data exfiltration from the analytical data store. Prompt injection attacks via AI-generated SQL.

Failure If Violated: Hallucinated SQL causes OOM crash on analytical store. Unbounded query runs consume system memory. Data store corrupted or queried for unintended data ranges.

Criticality: HIGH

Confidence: HIGH

Category: AIB, SEC, DAT

ACD-AIB-005

Constraint Statement: AI model outputs that produce sentiment or qualitative scores for use in position sizing must be converted to execution parameters through a validated, deterministic mathematical transformation. The mapping from AI output to position size must be fully specified and auditable — not inferred dynamically by the model itself.

Source Files: 05_adr_candidate_corpus (ADR-098), 05_root_constraint_registry (ADR-098)

Supporting ADRs: ADR-098

Problem Prevented: AI models producing unconstrained position size recommendations. Hallucinated Kelly fractions or sizing parameters submitted directly to execution without mathematical bounds.

Failure If Violated: AI directly determines position size with no deterministic bounds. Oversized positions submitted on AI sentiment misclassification. Capital overexposure from unvalidated AI-to-execution math.

Criticality: HIGH

Confidence: MEDIUM

Category: AIB, VAL, HUM

ACD-AIB-006

Constraint Statement: Any interface that connects an AI model to system tools or APIs must physically prevent execution-bound tool calls from being reachable by the AI component. Physical blocking is required — policy-level prohibition without architectural enforcement is insufficient.

Source Files: 05_adr_candidate_corpus (ADR-104), 05_root_constraint_registry (ADR-104)

Supporting ADRs: ADR-104, ADR-071

Problem Prevented: Prompt injection attacks that manipulate the AI component into calling execution-bound tools it is policy-prohibited from using. Social engineering via crafted inputs reaching execution paths.

Failure If Violated: Injected prompt overrides policy prohibition. AI component calls execution tool directly. Exchange order submitted without human gate via prompt injection.

Criticality: CRITICAL

Confidence: HIGH

Category: AIB, SEC, HUM

SECTION 8 — OPERATIONAL CONSTRAINTS

ACD-OPS-001

Constraint Statement: Data pipeline scheduled jobs must be designed to detect and prevent concurrent execution overlaps. Any pipeline task that writes to shared state must enforce mutual exclusion. Silent deadlock under API latency spikes must be treated as a testable, validated failure mode — not an assumed-safe condition.

Source Files: 05_adr_candidate_corpus (ADR-047, ADR-088, ADR-116), 09_architecture_readiness_reassessment

Supporting ADRs: ADR-047, ADR-088, ADR-116

Problem Prevented: Cron job overlaps creating deadlocks on shared data resources. API latency spikes causing job queue buildup and cascading deadlock.

Failure If Violated: Data pipeline deadlocks silently. Ingestion halts without alerting. Signal computation runs on stale data without detection.

Criticality: HIGH

Confidence: MEDIUM

Category: OPS, REL

ACD-OPS-002

Constraint Statement: Authentication token lifecycle management must be automated and must not require manual human intervention for routine token refresh cycles. Any authentication mechanism that requires daily manual action to maintain system connectivity is incompatible with unattended operation.

Source Files: 05_adr_candidate_corpus (ADR-087, ADR-117), 05_root_constraint_registry (ADR-087)

Supporting ADRs: ADR-087, ADR-117

Problem Prevented: System halt due to expired authentication token requiring manual refresh. Trading interruption from token expiry during unattended operating hours.

Failure If Violated: Authentication token expires during market hours. Trade execution daemon loses broker connectivity. Orders cannot be submitted. Positions cannot be managed.

Criticality: HIGH

Confidence: LOW (awaiting empirical validation of broker 2FA automation capability)

Category: OPS, REL, REG

ACD-OPS-003

Constraint Statement: The system must maintain a capital buffer explicitly reserved for the risk of retroactive exchange trade cancellations. This buffer is an operational requirement — it must be pre-allocated and must not be deployed as trading capital.

Source Files: 05_adr_candidate_corpus (ADR-114), 06_architecture_core (ADR-019)

Supporting ADRs: ADR-019, ADR-114

Problem Prevented: Margin calls triggered by retroactive clearinghouse cancellation of trades that were previously recorded as committed capital. Position state inconsistency between system records and cleared exchange state.

Failure If Violated: Clearinghouse voids trades the system considers committed. Net position diverges from exchange reality. Margin call issued against phantom committed capital.

Criticality: CRITICAL

Confidence: MEDIUM

Category: OPS, DAT, REG

SECTION 9 — SECURITY CONSTRAINTS

ACD-SEC-001

Constraint Statement: Any interface layer between an AI model and system tools must be treated as a prompt injection attack surface. Defense must be implemented at the architectural layer — tool availability to the AI component must be controlled by the system, not by the model's own policy compliance.

Source Files: 05_adr_candidate_corpus (ADR-104), 05_root_constraint_registry (ADR-104)

Supporting ADRs: ADR-104, ADR-071

Problem Prevented: Adversarial inputs injected via data feeds, document content, or crafted prompts that cause the AI component to take execution-bound actions.

Failure If Violated: Prompt injection overrides AI policy. Execution tools reachable via crafted input. Orders submitted via injection without human gate.

Criticality: CRITICAL

Confidence: HIGH

Category: SEC, AIB, HUM

ACD-SEC-002

Constraint Statement: All production access credentials, API keys, and authentication secrets must be managed through a controlled secrets management process. Credentials must not be embedded in source code, configuration files, or environment variables accessible to the application runtime without protection.

Source Files: 05_adr_candidate_corpus (ADR-008, ADR-087), 06_architecture_core (ADR-008)

Supporting ADRs: ADR-008, ADR-087

Problem Prevented: Credential exposure through source code repositories, logs, or runtime environment inspection. Unauthorized API access from leaked credentials.

Failure If Violated: Credentials exposed in repository or logs. Unauthorized access to broker API. Orders submitted by unauthorized actor. Regulatory breach from unauthorized session.

Criticality: CRITICAL

Confidence: HIGH

Category: SEC, REG

APPENDIX A — Constraint-to-Driver Traceability Matrix

Constraint ID	Architectural Driver	Driver Criticality
ACD-REG-001	DRV-007 Data Governance Standards	CRITICAL
ACD-REG-002	DRV-007 Data Governance Standards	CRITICAL
ACD-REG-003	DRV-007 Data Governance Standards	CRITICAL
ACD-REG-004	DRV-004 Multi-Source Feed Verification	HIGH
ACD-DAT-001	DRV-004 Multi-Source Feed Verification	HIGH
ACD-DAT-002	DRV-004 Multi-Source Feed Verification	HIGH
ACD-DAT-003	DRV-004 Multi-Source Feed Verification	HIGH
ACD-DAT-004	DRV-006 System Survivability Controls	CRITICAL

Constraint ID	Architectural Driver	Driver Criticality
ACD-DAT-005	DRV-001 Local First Data Ingestion	CRITICAL
ACD-DAT-006	DRV-001 Local First Data Ingestion	CRITICAL
ACD-DAT-007	DRV-004 Multi-Source Feed Verification	HIGH
ACD-DAT-008	DRV-004 Multi-Source Feed Verification	HIGH
ACD-DAT-009	DRV-005 Rigorous System Auditability	HIGH
ACD-VAL-001	DRV-001 Local First Data Ingestion	CRITICAL
ACD-VAL-002	DRV-001 Local First Data Ingestion	CRITICAL
ACD-VAL-003	DRV-005 Rigorous System Auditability	HIGH
ACD-VAL-004	DRV-005 Rigorous System Auditability	HIGH
ACD-VAL-005	DRV-005 Rigorous System Auditability	HIGH
ACD-VAL-006	DRV-006 System Survivability Controls	CRITICAL
ACD-VAL-007	DRV-006 System Survivability Controls	CRITICAL
ACD-REL-001	DRV-006 System Survivability Controls	CRITICAL
ACD-REL-002	DRV-006 System Survivability Controls	CRITICAL
ACD-REL-003	DRV-006 System Survivability Controls	CRITICAL
ACD-REL-004	DRV-006 System Survivability Controls	CRITICAL
ACD-REL-005	DRV-007 Data Governance Standards	CRITICAL
ACD-REL-006	DRV-003 Human Approval Gates	HIGH
ACD-GOV-001	DRV-005 Rigorous System Auditability	HIGH
ACD-GOV-002	DRV-003 Human Approval Gates	HIGH
ACD-GOV-003	DRV-005 Rigorous System Auditability	HIGH
ACD-HUM-001	DRV-002 Deterministic Execution Isolation	CRITICAL
ACD-HUM-002	DRV-003 Human Approval Gates	HIGH

Constraint ID	Architectural Driver	Driver Criticality
ACD-HUM-003	DRV-003 Human Approval Gates	HIGH
ACD-HUM-004	DRV-002 Deterministic Execution Isolation	CRITICAL
ACD-AIB-001	DRV-002 Deterministic Execution Isolation	CRITICAL
ACD-AIB-002	DRV-002 Deterministic Execution Isolation	CRITICAL
ACD-AIB-003	DRV-001 Local First Data Ingestion	CRITICAL
ACD-AIB-004	DRV-002 Deterministic Execution Isolation	CRITICAL
ACD-AIB-005	DRV-002 Deterministic Execution Isolation	CRITICAL
ACD-AIB-006	DRV-002 Deterministic Execution Isolation	CRITICAL
ACD-OPS-001	DRV-006 System Survivability Controls	CRITICAL
ACD-OPS-002	DRV-007 Data Governance Standards	CRITICAL
ACD-OPS-003	DRV-005 Rigorous System Auditability	HIGH
ACD-SEC-001	DRV-002 Deterministic Execution Isolation	CRITICAL
ACD-SEC-002	DRV-007 Data Governance Standards	CRITICAL

APPENDIX B — Outstanding Validation Blockers

The following constraints have LOW confidence due to unresolved empirical validation. These constraints are real and binding, but their exact threshold parameters require empirical proof before architecture can be finalized.

Constraint ID	Unresolved Parameter	Blocking ADRs	Required Validation
ACD-DAT-006	WAL recovery time under hard power-off	ADR-015, ADR-043	Force power-off test under live write load
ACD-DAT-009	Clearinghouse cancellation handling policy	ADR-019	Broker API documentation review + sandbox test
ACD-VAL-003	Concurrency limits at 50GB analytical data scale	ADR-023, ADR-119	Load benchmarking under production data volumes
ACD-REL-003	Broker API uptime during extreme market events	ADR-006, ADR-120	Empirical latency logging during tail events
ACD-OPS-002	OAuth 2FA automation capability with broker	ADR-117	Broker API sandbox test of token auto-refresh

APPENDIX C — Constraint Summary by Category

Category	Count	CRITICAL	HIGH	MEDIUM
Regulatory (REG)	4	3	1	0
Data (DAT)	9	4	5	0
Validation (VAL)	7	4	3	0
Reliability (REL)	6	3	3	0
Governance (GOV)	3	1	2	0
Human Oversight (HUM)	4	1	3	0
AI Boundary (AIB)	6	3	3	0
Operational (OPS)	3	1	2	0
Security (SEC)	2	2	0	0
TOTAL	**44**	**22**	**22**	**0**

ACD Version 1 — Constraint Discovery Phase — Pre-Architecture No architecture designed. No technologies recommended. No implementation planned. This document is a prerequisite input to architecture design, not an architecture artifact.